

Deep Learning-Based Selective Feature Fusion for Litchi Fruit Detection Using Multimodal UAV Sensor Measurements

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Abstract—In the field of precision agriculture, accurate crop detection is crucial for crop yield estimation, and health monitoring using photogrammetric measurements. Achieving high precision requires advance object detection models and multiscale feature fusion. This article addresses key research gaps in litchi crop monitoring, including the lack of a suitable dataset for litchi detection in natural environment and the limitations of conventional deep learning models in handling challenges such as occlusion, overlapping, and background complexities. First, we prepare high-resolution litchi dataset called “UAVLitchi” of 5000 images that include both RGB and multispectral images and next, we propose a selective feature fusion (SFF)-based architecture for litchi detection. By utilizing both RGB and multispectral images, this architecture effectively mitigates the challenges of visual detection arising from the complex cluster growth structure of litchis, offering a robust solution for accurate detection. The integration of SFF within a dual-channel mask-region based convolutional neural network (Mask-RCNN) leading to significant improvements in feature extraction for litchi detection. Experimental results demonstrate impressive performance, achieving an mean average precession (mAP50) of 94.65%, mAP75 of 89.23%, recall of 90.16%, and F1-score of 91.44%.

Impact Statement—This article makes a significant impact in the field of precision agriculture by addressing the critical challenges of fruit detection, specifically for litchis, using feature fusion techniques. By integrating the proposed selective feature fusion (SFF) model within the Mask-RCNN framework, this research significantly enhances the accuracy of litchi detection across different growth stages. This improvement is crucial for automating crop monitoring and yield estimation processes. The approach not only improves detection in complex scenarios involving occlusion and overlapping but also contributes to the broader adoption of multimodal sensing technologies in agriculture, paving the way for more precise and efficient crop management solutions. Additionally, the introduction of the UAVLitchi dataset for litchi detection and instance segmentation

can significantly benefit the research community in the field of precision agriculture.

Index Terms—Crop health monitoring, deep learning, feature fusion, object detection, precision agriculture, sensors, UAV.

I. INTRODUCTION

PRECISION agriculture, a rapidly growing field, is driving a surge of technological innovations to address challenges faced by farmers and decision-makers globally [1]. Accurate fruit detection, is a fundamental part of precision agriculture, is made possible through the integration of computer vision technology and advanced sensors. These innovations have opened up new domain for advanced research, enabling smart monitoring and efficient management of agricultural processes. Computer vision finds extensive applications in modern agricultural research, including dynamic vision-based servo control methods for continuous fruit harvesting operations [2], photogrammetric measurements [3] and robotic-based 3-D vision technology for external cracks damage recognition [4]. Moreover, accurate detection and consistently monitoring the growth of fruits like litchis is essential for automatic yield estimation. The unique clustered growth structure of litchi fruits pose a formidable challenge for its precise detection and boundary extraction [5]. It primarily challenges involves occlusion, overlapping and issues of uneven colour during its growth. Another major challenge is the detection of green litchis, especially during their early growth stages as they tend to blend in with the surrounding foliage (background complexity). Hence, it is difficult to detect and estimate boundary of individual instances of litchis using the existing conventional deep learning (DL) models [6] based on a single data modality [7]. Baseline models often struggle with challenges as shown in Fig. 1 for detection of crop in natural environment. Table I summarizes the key challenges faced by conventional DL models in fruit detection and highlights the unique strategies employed by the proposed methodology to address them effectively. Fortunately, the development of modern sensing technologies [8], DL approaches, particularly the multimodal feature fusion techniques [9], have ushered in a new era of possibilities across various domains, including medicine [10], industrial automation and agriculture [11]. This innovative approach involves the integration of information from multiple data sources or modalities to enhance the overall understanding and performance of the model. Among

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Fig. 1. Challenges in Litchi detection.

TABLE I
CHALLENGES IN CONVENTIONAL DL MODELS IN FRUIT DETECTION

SOTA DL Model for fruit detection	Challenges	Solution offered by the proposed model
YOLOV5 [14]	Occlusion and overlapping fruits	The selective feature fusion approach enables effective integration of features, enhancing the model's ability to distinguish fruit boundaries more accurately.
Faster-RCNN [15]	Small object detection	The proposed fusion model incorporates the Multi-Scale Channel Attention Module (MS-CAM), enhancing attention to small objects for improved detection.
Mask-RCNN+SE [16] Mask-RCNN+CBAM [3] YOLO-FT [17]	Background complexity	The multi-modal feature fusion from RGB and multispectral images mitigate the varying levels of background complexity by complementarily combining spatial and spectral features.

these technologies, DL and vision sensors are well-established and widely used in precision agriculture for tasks such as fruit detection and counting, as evidenced by numerous state-of-the-art (SOTA) research articles [11], [12], [13]. However, the integration of unmanned aerial vehicles (UAVs) and multimodal fusion techniques remains relatively underexplored. This leads to the need for a viable solution that can navigate the intricate complexities of segmentation of crops in a bunch. Therefore, in this article, we aim to address this gap by harnessing the potentials of UAV and multimodal sensors, focusing on the instance segmentation of litchi crops. The incorporation of RGB sensors allows us to leverage the rich colour information and visual cues, while multispectral sensors provide the spectral information, offering a comprehensive view of the crop growth. Additionally, we aim to design a lightweight feature fusion model that can be integrated with a mask region-based convolutional neural network (Mask-RCNN) [6] model to enhance the accuracy and robustness of litchi detection, while minimizing computational complexity. The core objective of this research is to explore and demonstrate the fusion of RGB and multispectral sensor images in conjunction with a modified Mask-RCNN model for effective detection of litchis at different growth stages for yield estimation.

The rest of the article is organized as follows: Section II provides a brief overview of related works. Section III gives the prior art. Section IV describes the data acquisition technique using UAVs and the datasets for the experiments. Section V includes the design of the proposed methodology and the rationale behind it. Section VI presents the experimental results, comparing the proposed model with state-of-the-art models and feature fusion architectures for multimodal data, using both UAV and non-UAV datasets. This section also includes a statistical analysis of the model's performance. Finally, Section VII concludes the article.

II. RELATED WORKS

Sa et al. [18] introduced near-infrared (NIR) and RGB datasets derived from public datasets such as nirsene and SEN12MS, as well as a novel dataset focused on sweet peppers. The sweet pepper dataset comprised 1615 NIR+RGB image pairs collected from commercial farms. Their work quantitatively and qualitatively demonstrates the suitability of these datasets for synthetic NIR image generation, achieving Frechet inception distances (FIDs) [19] of 11.36, 26.53, and 40.15 for nirsene, SEN12MS, and sweet pepper datasets, respectively. The model was evaluated using the YOLOv5 single-stage detector yields impressive mean-average-precision (mAP[0.5 to 0.95]) results ranging from 0.49 to 0.812. Back et al. [20] proposed a multimodal feature fusion approach employing domain randomization to prioritize shape over textures. After transferring the model to real-world settings, they compared its performance against baselines, carried a few ablation studies, showcasing its efficacy in segmenting industrial components with high clutters and occlusions [20] using purely synthetic data. Zeng et al. [21] introduced a novel feature fusion method called spatial pyramid pooling-balanced-feature pyramid network (ABFPN). Evaluated the model on three benchmark datasets, found the model efficient and reliable. It also enhances small object detection, improving printed circuit board (PCB) defect detection. Jiang et al. [22] introduced the Multiscale Feature Fusion Small Object Detection Network (MFFSODNet). To improve small object detection in UAV images, they replaced the large object prediction head with a tiny head, enhancing accuracy and reducing parameters. They also designed a multiscale feature extraction module (MSFEM) for fine-grained feature extraction. In addition, they proposed a bidirectional dense feature pyramid network (BDFPN) to fuse shallow and deep feature maps efficiently. Guo et al. [23] explored the potential of multispectral and hyperspectral images for estimating chlorophyll meter values, retrieving leaf area index (LAI), and predicting maize grain yields. Data were collected based on using two-band vegetation indices and textural indices. The results revealed that hyperspectral images, particularly their narrow bands, were more sensitive than multispectral images in areas of high vegetation coverage. The study concluded that integrating full hyperspectral spectra with random forest (RF) models is highly effective for accurate maize grain yield prediction. Yiming et al. [24] introduced a uncertainty-aware cross-modality vehicle detection (UA-CMDet) framework that enhanced vehicle detection in complex environments by using uncertainty-aware modules with cross-modal intersection over union (IoU) and illumination estimation. The integration of multimodal feature fusion with DL models has gained significant attention in the recent years, particularly within the realms of natural language Processing (NLP), bio-medical engineering and industrial automation [11]. Sun et al. [25] introduces a review on new methods for plant ecologists to address ecological questions from individual to regional scales using UAV remote sensing. Where it states that most studies emphasize data processing and remote sensing inversion techniques, ecological applications remain underexplored. Community-scale research dominated

61% of the studies reviewed, with RGB and multispectral cameras being the most commonly utilized sensors, accounting for 75% of usage. This approach offers new opportunities for ecological research advancement. However these concepts are widely used and established in these domains, there has been a noticeable gap in its application to precision agriculture [5], specifically in the sector of precise detection of small fruits in complex environments. Despite the extensive exploration of multimodal DL-architecture in other fields, the agricultural sector remains relatively unexplored in this regard. Further, the absence of a benchmark dataset to compare these models is a significant obstacle to develop robust DL architectures for smart agriculture. These research gaps motivated our investigation to design an optimal and high-precision litchi detection model. The proposed model aims to address contemporary challenges by considering environmental constraints and varying lighting conditions, which are major limitations for vision-based applications. It leverages advanced techniques in multimodal fusion to overcome these challenges effectively. The proposed model provide unique approaches such as deep selective fusion to mitigated occlusion and overlapping and enhance the model's ability to accurately distinguish fruit boundaries. Next, small object detection is improved by incorporating the multiscale channel attention module (MS-CAM). Finally, the challenge of blending of green litchi is addressed by multimodal fusion of features, which complementarily combine spatial and spectral information to manage varying background complexities. The major contributions of the article are listed below.

- 1) Proposed a multimodal UAVLitchi dataset for litchi detection and instance segmentation. The dataset can significantly contribute to the research community in the field of precision agriculture.
- 2) Proposed a simple but effective selective feature fusion (SFF) module to enhance small object detection. The SFF module consists of sub-networks integrated within the baseline Mask-RCNN architecture. The subnets are designed inspire based on concepts from the feature extraction architecture of the MS-CAM [9]. MS-CAM consists of unimodal feature extraction network, whose main function is to implement channel attention across different scales of feature maps by adjusting the kernel size and stride of the pooling layers. Further details of MS-CAM are provided in Section III. However, it is important to clarify that the MS-CAM is a single channel feature extraction model, whereas the proposed SFF is a dual-channel feature fusion model, which itself makes the proposed model an unique architecture for multimodal data fusion. The SFF model is specifically designed to selectively fuse features from multimodal data, effectively addressing the challenges of detecting litchi in a clustered growth environment.
- 3) Proposed a dual channel modified Mask-RCNN architecture to integrate the proposed SFF model.
- 4) Compared and statistically validated the proposed model with state-of-the-art (SOTA) instance segmentation models and provide a optimal DL model for litchi fruits boundary detection and estimation.

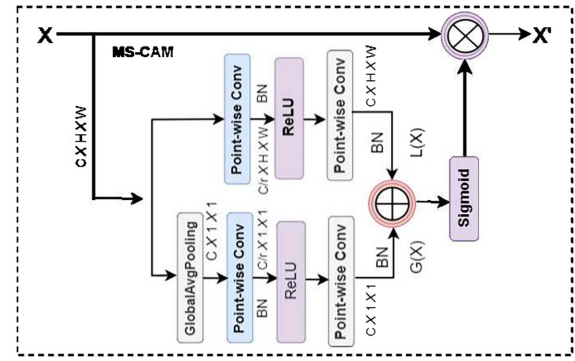


Fig. 2. Schematic diagram of MS-CAM.

III. PRIOR ART

A. Feature Extraction by MS-CAM

Fig. 2 illustrates the schematic diagram of the MS-CAM [9]. The main concept behind MS-CAM is that channel attention can be achieved at different scales of the same feature map by varying the kernel size and stride of the pooling layers. To keep the module as lightweight as possible, both local and global properties are incorporated within the attention block. Local content is aggregated using the point-wise convolution (PWConv) operation, while global context is aggregated using the global average pooling. These are then combined through interaction for each spatial position. The global content $G(X)$ and local content $L(X) \in \mathbb{R}^{C \times W \times H}$ are computed and saved as shown in the following equations [9]:

$$\begin{aligned} L(X) &= BN(PWConv_2(\delta(BN(PWConv_1(X))))) \quad (1) \\ G(X) &= BN(PWConv_2(\delta(BN(PWConv_1(GAP(X))))) \quad (2) \end{aligned}$$

The kernel size of $PWConv_1$ and $PWConv_2$ are $(C/r) \times C \times 1 \times 1$ and $C \times (C/r) \times 1 \times 1$, respectively. Given the $L(X)$ and $G(X)$, the refined feature $\hat{X} \in \mathbb{R}^{C \times H \times W}$ can be obtained as shown in the following equation [9], where $M(X) \in \mathbb{R}^{C \times W \times H}$ denote:

$$\hat{X} = X \otimes M(X) = X \otimes \sigma(L(X) \oplus G(X)) \quad (3)$$

the refined weights, $\oplus$ and $\otimes$ denotes the broadcasting addition and element-wise multiplication operation, respectively. δ is the ReLU and σ is the sigmoid activation function. BN is the batch normalization layer, GAP is the global average pooling layer.

B. Attention Feature Fusion (AFF)

Feature fusion across multimodal data are considered as one of the challenging problems in small object detection [26], which can be described mathematically in a DL framework as

$$F_{Fused} = f(F_R, F_M) = f(\phi(I_R), \phi(I_M)) \quad (4)$$

where I_R and I_M represent inputs from RGB and multispectral sensors, respectively. They produce RGB feature maps F_R and multispectral feature maps F_M using feature extraction functions $\phi_R(\cdot)$ and $\phi_M(\cdot)$, respectively. The final fused feature map

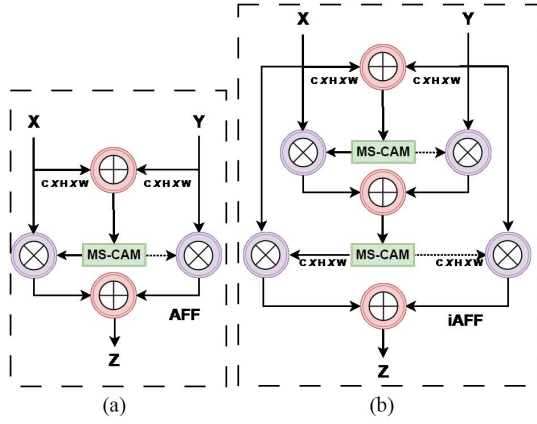


Fig. 3. Schematic diagram of: (a) AFF and (b) iAFF.

F_{Fused} is created using the fusion function $f(\cdot)$ as shown in (4) [26]. Based on the above philosophy of multimodal fusion, author in [9] introduce a multimodal feature fusion network called AFF. AFF is designed by integrating the MS-CAM feature extraction module as shown in Fig. 3(a). AFF takes two features maps X and $Y \in \mathbb{R}^{C \times W \times H}$ as an input, where Y is the feature map with a larger receptive field. X is the output of a 3×3 kernel, and Y is the output of a 5×5 kernel, as seen in InceptionNet. In the short skip connection scenario, X represents the identity mapping, while Y is the learned residual within a ResNet block. Therefore, the fused feature Z in AFF can be represented as

$$Z = M(X \uplus Y) \otimes X + (1 - M(X \uplus Y)) \otimes Y. \quad (5)$$

But with AFF the author in [9] expressed their concerned that the initial combination of features can significantly impact the final outcome, particularly in fully context-aware models. To address this issue, they proposed an another fusion network called iterative AFF (iAFF), which is an extended version of AFF. The iAFF network is designed in a two-stage configuration as shown in Fig. 3(b), with additional attention applied at each stage, ensuring a more refined and effective integration of features. Now, the initial integration of $X \uplus Y$ can be reformulated as

$$X \uplus Y = M(X + Y) \otimes X + (1 - M(X + Y)) \otimes Y. \quad (6)$$

Summary: The iAFF is an effective feature fusion model that performs well in small object detection and addresses challenges encountered during the initial integration of inputs. While it excels in detecting small objects, its performance declines as the complexity of the data increases. This is evident in the UAVLitchi dataset, which features small litchi fruits along with challenges such as overlapping, occlusion, and uneven color. The iAFF model struggled to maintain consistency in this scenario due to its rigid nature, as it does not fuse different scales of feature maps selectively. Moreover, due to its iterative two-stage configuration and its integration in the entire backbone networks of Mask-RCNN architecture makes it a computationally heavy. Therefore, we proposed the SFF model, which is lightweight, flexible, and, most importantly it offer

SFF. In SFF, low-level features are fused with a local fusion subnet to enhance boundary detection, while high-level features are fused with a global fusion subnet for effective contextual understanding of the cluster growth structure of litchi. A detailed description of the proposed SFF model and the modified Mask-RCNN architecture is provided in Section V.

IV. MATERIALS AND METHODS

The scarcity of UAV-sensed crop datasets, in particular, for litchi fruit detection, pose a major challenge in developing a robust DL model. To address this, we propose a UAV-based multimodal litchi dataset (UAVLitchi) across various energy bands: RGB, Red, Green, NIR, and RedEdge. We use DJI Phantom 4 UAV equipped with in-built RGB sensor 20MP (5472×3648). Additionally, the drone is integrated with a Parrot Sequoia multispectral sensor to collect multispectral images across the four bands. Detailed description of the sensors are as follows: 1) CMOS sensor: It is an 1-inch 20 MP sensor with a mechanical shutter that eliminates rolling shutter distortion, with 94° FOV and fixed lens of focal length 8.6 mm. 2) Parrot Sequoia multispectral sensor: It is capable of measuring images in four different multispectral bands with a resolution of 1.2 MP (1280×960 pixel) for each band. The multispectral bands include Green ($550 \text{ nm} \pm 40 \text{ nm}$), Red ($660 \text{ nm} \pm 40 \text{ nm}$), Red edge ($735 \text{ nm} \pm 10 \text{ nm}$), and Near infrared ($790 \text{ nm} \pm 40 \text{ nm}$), providing detailed data for crop analysis.

A. Data Acquisition and Dataset Preparation

UAVLitchi data are acquired from a litchi garden situated at Tezpur having geographical co-ordinates of $26^\circ 40' 49.7''\text{N}$ $92^\circ 48' 52.3''\text{E}$, approximately 5 km away from Tezpur University. The flight campaign was carried out from 5 March to 31 May 2024. We began by selecting a sample area containing 15 trees, arranged in three rows, each tree is of approximately equal height and at consistent spacing. High-resolution RGB images were captured using a DJI Phantom 4 drone equipped with a CMOS sensor with a 3:2 aspect ratio and an image resolution of 5472×3648 pixels. Given the average height of the litchi trees, which is 7 m, the drone's flight altitude was set at 8 meters above the ground, having the field of view (FoV) of the camera 84° horizontally and 63.7° vertically, facilitating optimal image coverage. The UAV capture images at three different shooting angles: 90° , 45° to the right, and 45° to the left. Whereas, the multispectral data acquisition were carried out using Parrot Sequoia multispectral sensor at the same height in timelapse mode with a time interval of 1.5 sec. In a single flight, we captured 500 images, of which 200 were using the CMOS sensor and the remaining 300 with the multispectral sensor. In total, we conducted six flight campaigns, amassing 3000 images from various locations within the sample area, at different angles and under varying lighting conditions to incorporate more realistic data distribution. As the data acquisition encompassed the entire litchi harvesting period, we were able to capture the litchis in every stages of growth: (immature, mature, colour break, and ripening stage). It includes litchis of varying colors during the growth phase, resulting the dataset both diverse and rich.

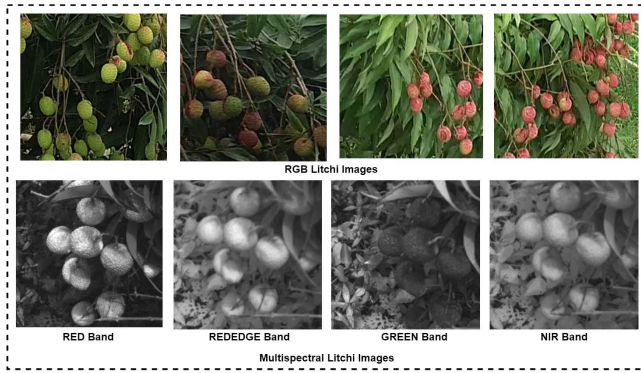


Fig. 4. Sample images from UAVLitchi (RGB and multispectral).

The default images are of size 4608×3456 and 1280×960 for RGB and multispectral bands, respectively. To lessen the computational load while training, we converted the raw images into smaller image samples of size 256×256 as shown in Fig. 4. Finally, we obtained a refined dataset of 5000 images ready for annotation after data pruning.

B. Data Preprocessing and Augmentation

Subsequently, the UAV images were normalized for pixel intensity, where the pixel values of both image types were scaled to a common range. We accomplish this using min-max scaling [27]. This step is primarily executed to guarantee that both modalities of data contribute proportionally during the training phase. Then we utilize Labellmg [5] tool, to perform the litchi annotations, resulting in approximately 60 000 annotations. The annotations were saved in an XML file using the PASCAL VOC [28] format. Out of 5000 images, 70% data are used for training, while 20% for validation and 10% are kept for testing. Further, To provide a more accurate estimate of the model performance on unseen data, we also augment the training set using several techniques. These include geometric rotations of 90° and 180° , addition of Gaussian noise, and color transformations, such as RGB color intensification and image sharpening. As a result, the training set increased to 8500 images, with 1000 images used for validation and the remaining 500 images for testing, bringing the total dataset to 10 000 images.

V. PROPOSED METHODOLOGY

The proposed method involves designing and integrating a feature fusion model called SFF with the baseline Mask-RCNN architecture. SFF comprises of five sub-blocks: two local-local feature fusion (LLFF), two global-global feature fusion (GGFF), and one global-local feature fusion (GLFF). Each of the SFF's sub-blocks are explained in detail below.

Local-Local Feature Fusion (LLFF): The LLFF design combines the local features from two different modalities of images: RGB and multispectral. This is achieved within a single attention module similar to MS-CAM. LLFF consists of two bottleneck networks that are symmetrical in nature, as shown in Fig. 5(a).

Let X and $Y \in \mathbb{R}^{C \times W \times H}$ be the two feature maps having the same respective field. X and Y are the outputs of a 3×3 kernel in the ResNet50 backbone network. Furthermore, both the feature maps consist of identity mapping and learned residual layer. $L(X)$ and $L(Y)$ are the local features aggregated from RGB and multispectral images, and the fused featured map, respectively as represented by the the following equations:

$$\begin{aligned} L(X) &= L(Y) \\ &= BN(PWConv_2(\delta(BN(PWConv_2(X \uplus Y))))), \end{aligned} \quad (7)$$

and

$$Z = [(X \otimes \sigma(L(X) \oplus L(Y))) \oplus (Y \otimes (1 - \sigma(L(X) \oplus L(Y))))] \quad (8)$$

where M denotes the selective weights, Z is the fused feature map generated from X and Y . δ is the Laeky ReLU activation function and $\uplus$ denote the initial integration of two feature X and Y .

Global-Global Feature Fusion (GGFF): GGFF is similar to the LLFF with one additional layer of global average pooling layer (GAP) in both the bottleneck branches as shown in Fig. 5(b). It primarily focuses on the global content and fuse the global feature maps from both the modalities. The mathematical representation for the fused feature map Z for GGFF is shown in the following equation: The key distinction between LLFF and GGFF arises from the integration of the GAP layer. The global feature extractors from the bottleneck structure are represented by $G(X)$ and $G(Y)$, with its global parameter calculated as in the following equation:

$$Z = [(X \otimes \sigma(G(X) \oplus G(Y))) \oplus (Y \otimes (1 - \sigma(G(X) \oplus G(Y))))] \quad (9)$$

$$\begin{aligned} G(X) &= G(Y) \\ &= BN(PWConv_2(\delta(BN(PWConv_2(GAP(X \uplus Y)))))) \end{aligned} \quad (10)$$

Global-Local Feature Fusion (GLFF): The GLFF is the final sub-block within the SFF module. This block is designed to extract global features from RGB images and local features from multispectral images. It consists of one global context aggregator and one local context aggregator bottleneck network, as shown in Fig. 5(c). GLFF is specifically designed to focus on the intermediate feature map of ResNet50, which contains both local and global features in approximate equal proportions. The fused feature map is represented by Z , as shown in the following equation:

$$Z = [(X \otimes \sigma(G(X) \oplus L(Y))) \oplus (Y \otimes (1 - \sigma(G(X) \oplus L(Y))))] \quad (11)$$

Furthermore, the global feature aggregator $G(X)$ and the local feature aggregator $L(Y)$ for GLFF remain the same as depicted in (7) and (10), respectively. With the above architectural concepts on attention, we can now explain the proposed model as follows:

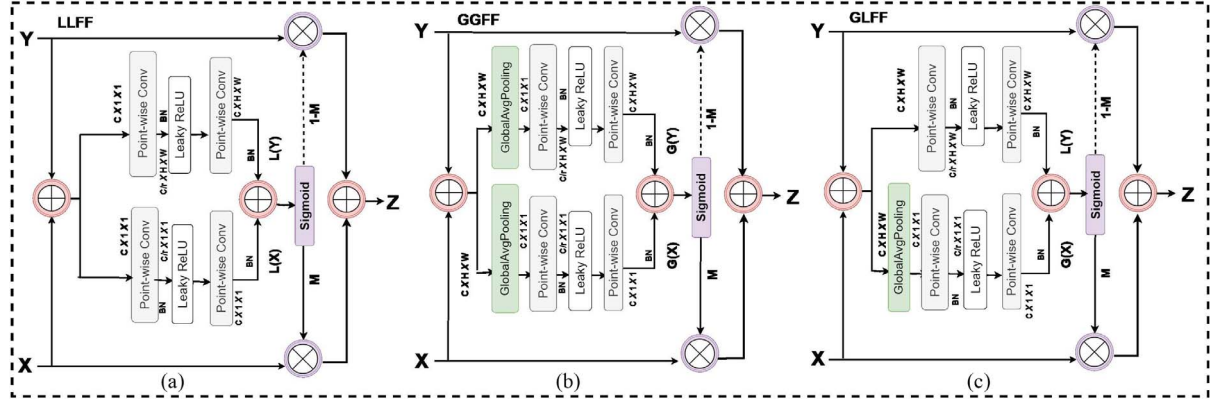


Fig. 5. Schematic diagram of subnet inside SFF: (a) Local Local Feature Fusion (LLFF); (b) Global Global Feature Fusion (GGFF); and (c) Global Local Feature Fusion (GLFF).

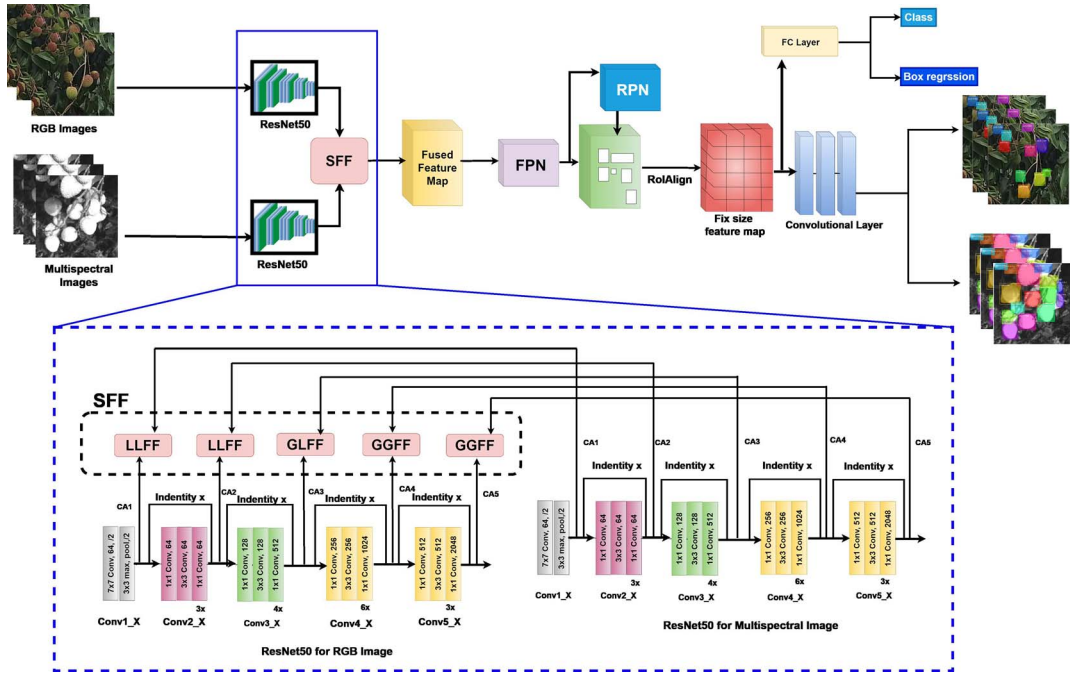


Fig. 6. Complete block diagram of the proposed DL method: SFF serves as a multimodal feature fusion block, feature pyramid network (FPN) acts as a second-level feature extractor, converting input features of varying sizes into a unified single scale and the region proposal network (RPN) refines the bounding box coordinates to enhance detection accuracy.

A. SFF-based Dual Channel Mask-RCNN Architecture for Litchi Segmentation

Fig. 6 illustrates the complete block diagram of the SFF-based dual-channel Mask-RCNN DL architecture designed for litchi segmentation. The proposed SFF module serves as a multimodal feature fusion block, seamlessly integrated with the ResNet50 backbone network, while the FPN acts as a second-level feature extractor, converting input features of varying sizes into a unified single scale. Additionally, the RPN refines the bounding box coordinates to enhance detection accuracy. The baseline Mask-RCNN architecture has been modified to a dual-channel configuration to simultaneously train RGB and multispectral images with ResNet50 baseline architecture. Each

TABLE II
TRAINING AND HYPER-PARAMETERS DETAILS; STOCHASTIC GRADIENT DESCENT (SGD)

Training parameter		Hyperparameter	
Backbone Network	ResNet50	Learning rate	0.001
Kernel Size	3×3	Weight delay	0.0001
Filter Size	2×2	Batch size	32
Backbone Strides	[4,8,16,32,64]	Epochs	200
RPN Anchor Scale	[32,64,128,256,512]	Optimizer	SGD

ResNet50 consists of five stages, corresponding to five distinct feature map scales (CA1, CA2, CA3, CA4, CA5). Given the varying scales of feature maps, we design a feature fusion model that selectively fuses features from every stages of ResNet50.

TABLE III
ABLATION STUDY OF THE PROPOSED METHOD USING UAVLITCHI DATASET IN TERMS OF GIGA FLOATING POINT OPERATIONS PER SECOND (GFLOPs), MAP50, MAP75, RECALL AND F1-SCORE AT INTERSECTION OVER UNION (IoU) = 0.75, COM. STANDS FOR COMBINATION

Combinations of SFF	Feature Maps					GFLOPs	mAP50 (%)	mAP75 (%)	Recall (%)	F1_Score (%)
	CA1	CA2	CA3	CA4	CA5					
Com. 1 GGFF	✓	✓	✓	✓	✓	120.8	90.52	81.98	84.34	82.88
Com. 2 LLFF	✓	✓	✓	✓	✓	108.1	86.53	85.98	83.76	84.98
Com. 3 GLFF	✓	✓	✓	✓	✓	117.2	83.99	78.59	80.21	79.36
Com. 4 LLFF GLFF	✓ ✗	✓ ✗	✗ ✓	✗ ✓	✗ ✓	110.3	85.25	79.87	81.67	80.75
Com. 5 LLFF GLFF GGFF	✓ ✗ ✗	✓ ✗ ✗	✗ ✓ ✗	✗ ✓ ✓	✗ ✗ ✓	112.4	94.65	89.23	90.16	91.44

TABLE IV
PERFORMANCE COMPARISON ON DIFFERENT DATASETS AND SOTA FUSION MODELS IN TERMS OF MAP75, RECALL AND F1-SCORE AT INTERSECTION OVER UNION (IoU) = 0.75

Dataset	Modality	Methods	mAP75 (%)	Recall%	F1_Score%
DeepNIR Capsicum [18]	RGB + NIR	YOLOV5 + CFT	84.90	85.54	83.65
		Faster-RCNN + iAFF	87.88	89.98	88.53
		YOLOV5 + GFF	85.76	86.11	84.68
		Proposed Method	89.62	90.87	89.34
LLVIP [30]	RGB+ IR	YOLOV5 + CFT	87.21	89.23	90.11
		Faster-RCNN + iAFF	77.93	79.87	80.98
		YOLOV5 + GFF	75.52	76.90	75.89
		Proposed Method	89.95	91.89	93.12
DroneVehicle [24]	RGB+ IR	YOLOV5 + CFT	84.65	80.43	81.98
		Faster-RCNN + iAFF	87.64	89.56	90.76
		YOLOV5 + GFF	85.94	86.67	87.77
		Proposed Method	89.98	91.88	92.89
UAVLitchi	RGB + Multispectral	YOLOV5 + CFT	85.98	83.32	84.32
		Faster-RCNN + iAFF	87.11	89.98	88.65
		YOLOV5 + GFF	86.81	88.65	84.32
		Proposed Method	89.23	90.16	91.44

The sub-blocks inside the SFF block are placed systematically so that the first two feature maps from the ResNet50 network (CA1 and CA2) are fused using the LLFF block. The middle feature map (CA3) is fused using the GLFF block, and the final two feature maps (CA4 and CA5) are fused using the GGFF block. The LLFF block is connected to the initial feature maps due to their low-level, local feature contents while the GGFF block is connected to the final feature maps because of their rich representations of global, high-level features. Deka et al. [16] have already reported a design of a modified Mask-RCNN architecture on a similar hypothesis by integrating a channel attention module selectively within the backbone along with the FPN for litchi detection. After achieving the overall feature enhancement, the feature maps are fed into the RPN [29], which generates potential bounding box locations for objects in an image. At each position, the RPN predicts the presence of an object, refines the bounding box coordinates, and assigns a probability score to each proposal based on the likelihood of containing an object. These proposals are ranked, and the top-scoring regions-of-interest (RoIs) are selected for further processing, including bounding box refinement and instance segmentation, to produce the final object detection and mask prediction. Integrating the SFF block with the baseline Mask-RCNN architecture and adopting a multimodal approach has yielded a robust segmentation model for litchi detection in complex natural environments. The SFF module, comprising

five sub-blocks, effectively fuses local and global features from RGB and multispectral images, enhancing detection accuracy. Additionally, the integration of MSCAM improves detection on small objects. By systematically integrating the SFF module within the backbone network, the proposed model effectively identifies key features across various litchi growth stages. It further addresses the challenges posed by background complexity through a multimodal selective fusion approach that complementarily combines spatial and spectral features. This innovative design significantly outperforms existing DL methods in both accuracy and precision.

VI. EXPERIMENTS AND RESULTS

A. Experimental Setup

Simulations were conducted using the Spyder IDE with Python 3.7.12 on a general-purpose graphics processing units (GPGPU) server equipped with 32GB RAM Tesla V100 and a drive storage of 1TB. All DL models are evaluated with SOTA object detection metrics which includes: mean Average Precisions (mAP50) and mAP75, Recall and F1 Score. Here, mAP50 is computed at Intersection over Union (IoU) = 0.50 and mAP75 is calculated at IoU = 0.75 [26]. Table II illustrate the training parameter and hyperparameter in details.

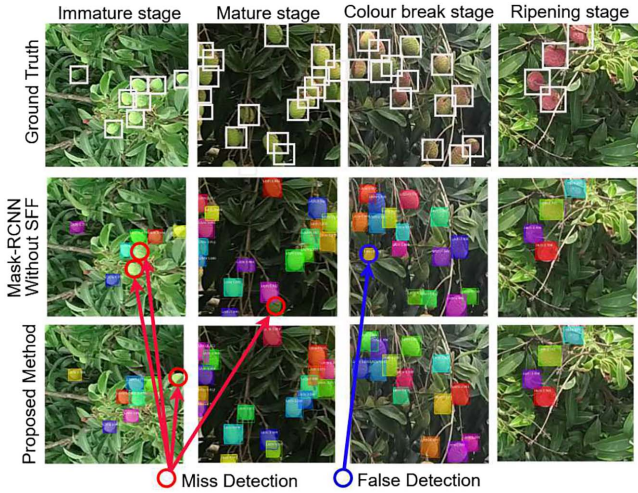


Fig. 7. Visual comparison results at different stages of litchi growth with and without SFF module in term of miss and false detection.

B. Ablation Study

In the ablation study, we conducted five sets of experiments on the UAVLitchi dataset, evaluating the proposed model with various combinations of fusion sub-blocks as shown in Table III, within the SFF model. These sub-blocks were employed to fuse the five scales of feature maps (CA1, CA2, CA3, CA4, and CA5) generated by the backbone networks from different modalities. The results demonstrated that fusing only with the GGFF produced good detection results but required higher giga floating point operations per Second (GFLOPs) compared to the proposed model. Conversely, using only the LLFF resulted in lower accuracy but required fewer GFLOPs. But we observed a significant improvement in detection performance when all three sub-blocks were combined and selectively placed within the architecture. We strategically integrate the sub-blocks by fusing low-level features (CA1 and CA2) with the LLFF, intermediate features (CA3) with the GLFF, and high-level features (CA4 and CA5) with GGFF. The proposed selective fusion achieved the highest mAP50 and mAP75 of 94.65 and 89.23%, respectively, with the GFLOPs of 112.4. This approach not only improves model accuracy but also proves to be the most efficient feature fusion architecture in terms of both computational cost and detection performance.

C. Comparison Study

Table IV compares the proposed model with the SOTA detection models across various benchmark datasets, including the DeepNIRCapsicum [18], the Visible-Infrared Paired Dataset for Low-Light Vision (LLVIP) [30], and the DroneVehicle [24]. Fig. 9 presents a comparison of visual results obtained using different models on the UAVLitchi dataset. The comparison highlights performance in various challenging scenarios, including green litchis (GL), litchis with uneven colour (LU), and litchis with overlapping and occlusions

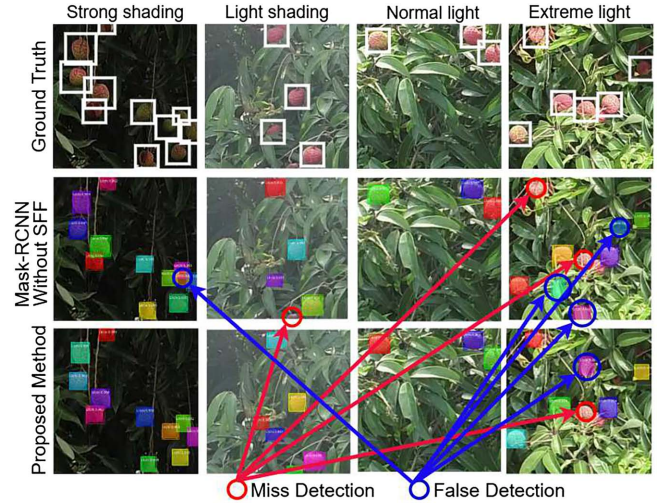


Fig. 8. Visual comparison results at different lighting conditions with and without SFF module in term of miss and false detection.

(LO). Figs. 7 and 8 illustrate the visual detection results of the proposed model across various growth stages and lighting conditions.

Table V presents a comparison of the proposed model with the SOTA fusion models in terms of the number of parameters and inference time on the UAVLitchi dataset. This comparison highlights that the proposed model is computationally lighter compared to the other models such as the iAFF, the CFT, the GFF, the MSDS-RCNN and the AR-CNN. Fig. 10 displays the confidence interval, where the highest confidence interval (CI) score belongs to the GL class, while the lowest corresponds to the LU class. Additionally, the performance of the proposed model over different classes (GL, LU, OL) is shown in Fig. 11, illustrates the correlation with various SOTA feature fusion models compared to the proposed SFF model. The results indicate that the SFF model consistently outperforms all other feature fusion models across all classes. From these results, we can infer that the proposed model is capable of detecting litchis in all challenging cases and outperforms all the SOTA fusion models. This demonstrates the robustness and efficacy of the proposed DL model for litchi counting, highlighting its potential use in litchi yield estimation. To further demonstrate the effectiveness of the proposed model, additional results are included in the supplementary material. Table SI examines the performance of various state-of-the-art (SOTA) feature fusion modules within the same object detection framework, while Table SII evaluates the proposed SFF module across different object detection models. The results from both tables clearly demonstrate that the SFF module significantly enhances the detection performance of the Mask-RCNN by effectively integrating the multimodality features. Additionally, Fig. S1 illustrates the receiver operating characteristic curves, highlighting the trade-offs between true positive and false positive rates. Fig. S2 showcases a visual detection result on multispectral images, further validating the module's performance.

TABLE V
COMPARISON PERFORMANCE OF THE PROPOSED FUSION METHOD WITH STATE-OF-THE-ART MODELS USING UAVLITCHI DATASET IN TERMS OF NO. OF PARAMETERS, INFERENCE T, MAP50, RECALL AND F1-SCORE AT INTERSECTION OVER UNION (IoU) = C0.5

Modality	Fusion Model	Detector	No. of Parameters	Inference time (Sec)	mAP50 (%)	Recall (%)	F1_Score (%)
RGB	-	Mask-RCNN	55.5 M	7.21	85.73	85.98	86.99
Multispectral	-	Mask-RCNN	55.5 M	7.00	84.08	86.56	85.26
	CFT [31]	YOLOV5	88 M	7.74	91.76	85.43	87.22
	iAFF [9]	Faster-RCNN	135 M	8.32	93.01	88.67	88.87
	GFF [32]	YOLOV5	96 M	7.75	91.55	86.98	87.44
RGB + Multispectral	MSDS-RCNN [33]	Mask-RCNN	90 M	7.89	89.08	82.44	82.55
	AR-CNN [34]	Fast-RCNN	95 M	7.66	88.89	81.87	79.98
	Halfway Fusion (HF) [35]	Faster-RCNN	75 M	7.31	81.95	80.98	81.87
	SFF	Mask-RCNN	79 M	7.41	94.65	92.62	93.44
	Proposed Model	Mask-RCNN	79 M	7.41	94.65	92.62	93.44



Fig. 9. Detected Litchis using SOTA models on images from UAVLitchi dataset.

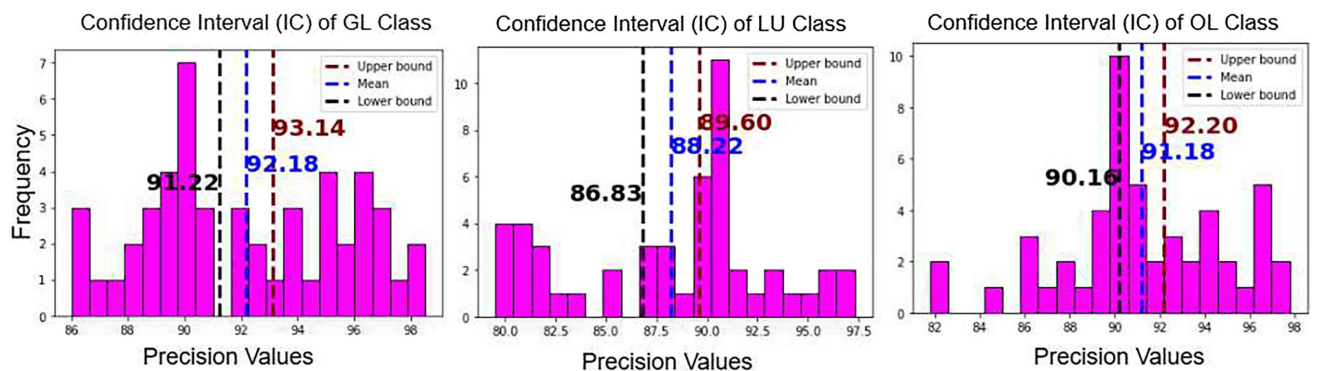


Fig. 10. Confidence Interval curve for different classes.

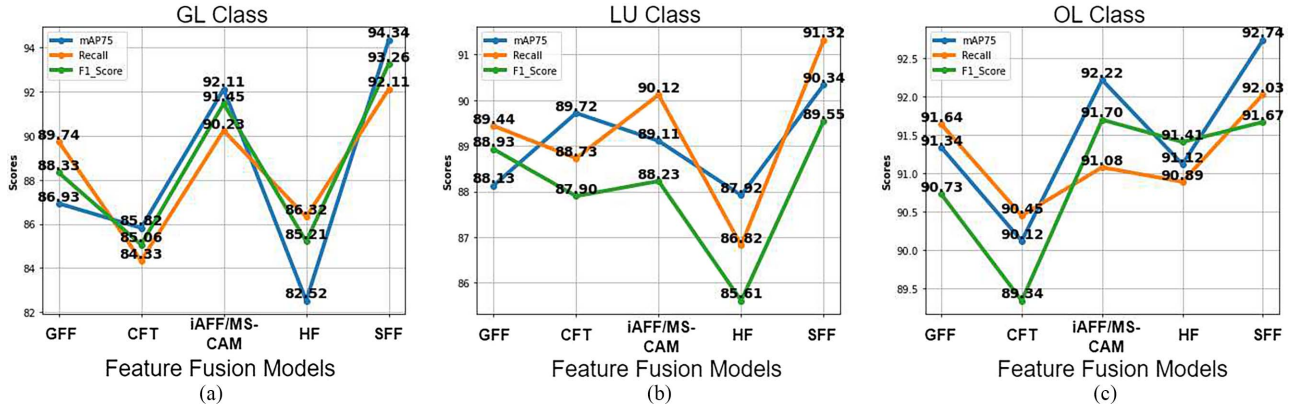


Fig. 11. Data correlation graph: (a) GL class w.r.t feature fusion models; (b) LU class w.r.t feature fusion models; and (c) LO class w.r.t feature fusion models.

VII. CONCLUSION

In this work, we proposed a novel dual channel Mask-RCNN model-based on SFF for litchi detection and boundary extraction. In addition, we constructed a multimodality UAVLitchi dataset consisting of both RGB and multispectral images of litchis. The model has been trained and validated using both benchmark datasets and the proposed UAVLitchi dataset. The results demonstrate its robustness across different data sources. The performance of the model was evaluated against widely used SOTA fusion models, comparing the number of parameters and inference time. The results of the study indicate that the proposed model demonstrates improved detection accuracy with minimum numbers of parameters of 79 M, and achieving the second best inference time of 7.41 sec. Furthermore, the experimental results reveal that the proposed model achieves an mAP50, mAP75, recall, and F1-score of 94.65, 89.23, 90.16, and 91.44%, respectively. While the model is absolutely scalable to other perennial crops like mango, citrus, orange, and apple, but the proposed model's performance also depends on adhering to specific UAV flight protocols, such as optimal flight height and shooting angles, flight speed, setting appropriate ground sampling distance during data collection tailored to the crops characteristics. We recognize certain limitations of the proposed model, including miss and false detections under a few specific conditions: 1) during the immature stages of litchi growth; and 2) under extreme lighting conditions, the former may be due to high resemblance of litchi with the background and at high light intensity boundary colour difference of the object cannot be perceived due to the sensor saturation. To address these challenges, our future work will include incorporating a sunshine sensor during data acquisition to normalize light intensity under extreme light conditions. In addition we also aim to develop a preharvest litchi yield prediction model. This model will integrate various components for comprehensive monitoring of litchi crops including health monitoring using vegetation indices, photogrammetric crown area estimation, and the proposed detection model as integral parts for litchi detection and counting to optimize litchi cultivation practices and yield prediction as an application of precision agriculture.

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